João Eduardo Pereira dos Santos

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ABOUT ME

Senior Python Developer with 16+ years of software engineering experience, currently expanding my expertise into AI Engineering and LLM-based applications. Strong background in Python, REST APIs, data integration, SQL, cloud and enterprise systems, with hands-on learning and practical exploration of LLMs, RAG, embeddings, prompt engineering and AI agents. Experienced in translating business requirements into scalable backend solutions and integrating enterprise systems through APIs.

WORK EXPERIENCE

PYTHON FULL STACK DEVELOPER – CTW - CRITICAL TECH WORKS – 14/11/2025 – Current – PORTO, PORTUGAL

- Develop and integrate Python-based backend services, REST APIs and enterprise applications, with growing focus on AI-enabled solutions and automation.
- Design and implementation of backend services and frontend components for enterprise applications
- Design and execution of data integration solutions for complex enterprise systems
- Extensive use of SQL for data extraction, transformation, integration, and reporting
- Development and integration of APIs to connect enterprise systems and services
- Containerization of applications using Docker to support standardized deployments
- Contribution to DevOps workflows, including automated builds, testing, and deployment pipelines
- Support and maintenance of applications and services running in Kubernetes and AWS environments
- Source code management and collaboration using Git
- Close collaboration with cross-functional teams in an Agile/Scrum environment

PYTHON FULL STACK DEVELOPER – EY - ERNST & YOUNG – 03/12/2022 – 15/11/2025 – BRAZIL

- Developed a Python-based application exploring **Retrieval-Augmented Generation (RAG)** for question answering over private documents
- Performed document ingestion, chunking, embeddings and semantic retrieval
- Integrated an LLM to generate contextual answers based on retrieved information
- Developed full-stack solutions, APIs, and data integration services using Python and JavaScript
- Participated in SAP migration projects, supporting data extraction, transformation, validation, and integration
- Developed and maintained data integration workflows using IBM DataStage
- Worked extensively with SAP tables and enterprise data, analyzing, extracting, transforming, and validating business data
- Developed Python scripts and SQL solutions to automate data processing, migration, and validation activities
- Designed and implemented integrations between SAP, databases, DataStage, and other enterprise systems
- Developed and supported dashboards and reporting solutions to provide business and operational insights
- Led and collaborated with technical teams in the delivery of complex data integration and API projects
- Performed technical analysis, troubleshooting, and data validation across enterprise environments
- Collaborated with business and technical stakeholders in an Agile environment

PYTHON FULL STACK DEVELOPER – UNIVERSIDADE ESTADUAL DE CAMPINAS – 03/04/2008 – 05/12/2022 – CAMPINAS, BRAZIL

- Developed full-stack web applications using Python, Flask, Django, JavaScript, and SQL
- Designed and developed RESTful APIs and backend services
- Developed frontend functionality and integrations for academic and administrative systems
- Designed and optimized relational databases using PostgreSQL, MySQL, and Oracle
- Led digital transformation initiatives across institutional systems
- Automated deployments and system maintenance using DevOps practices
- Managed Linux servers and application environments
- Developed BI solutions and automated workflows for large-scale datasets
- Provided technical guidance on architecture, technologies, and development practices

DIGITAL GAMES PROGRAMMING PROFESSOR – SESC SANTANA – 28/05/2016 – 28/08/2019 – BRAZIL

Game Programming: Taught students the fundamentals and advanced concepts of game programming, covering languages such as Lua, Python, and using engines like Blender Engine.

SOFTWARE ANALYST – MERCER BRASIL – 29/12/2005 – 12/03/2008 – SÃO PAULO, BRAZIL

Database Integration: Developed and implemented database integration solutions using SQL Server, ensuring efficient and secure data transfer between systems.

Retirement Simulators: Created retirement simulators in ASP, enabling users to project retirement benefits based on various scenarios with accurate and reliable results.

System Maintenance: Conducted ongoing maintenance and updates, fixing bugs, enhancing features, and ensuring compatibility with new technologies and legal requirements.

INTERNET SUPPORT SPECIALIST – ATENTO – 01/04/2001 – 01/02/2002 – SAO PAULO, BRAZIL

User Assistance: Provided technical support to users of Terra and Telefonica, resolving connectivity issues and troubleshooting errors to ensure seamless internet service.

Problem Resolution: Diagnosed and resolved a wide range of technical issues, including network configurations, software problems, and hardware malfunctions.

EDUCATION AND TRAINING

23/07/2003 – 27/12/2007 São Paulo, Brazil
DATA PROCESSING TECHNOLOGIST Faculdade de Tecnologia de São Paulo

Website http://www.fatecsp.br/

28/12/2019 – CURRENT Brazil
COMPUTER ENGINEERING Universidade Virtual do Estado de São Paulo

Website www.univesp.br

06/01/2023 – 30/01/2023 United States
BUILDING WEB APPLICATIONS IN DJANGO University of Michigan

Website https://umich.edu/

16/05/2023 – 24/07/2023 Rio de Janeiro, Brazil
ACCREDITED - FULL STACK BASIC Pontifícia Universidade Católica do Rio de Janeiro

Website https://pucrio.grupoa.education/academic-services/documents/validate/1c0115d0

LANGUAGE SKILLS

Mother tongue(s): **PORTUGUESE**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C2	C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Python | Pentaho | AppsScript | SQL | Scrum | Wordpress | Linux | JavaScript | Oracle | Git | MySQL | Microsoft Office | Visual Studio Code | RESTful Web Services | Devops: Docker, Jenkins | Back-end (Flask,

Django) | API Python frameworks: Flask, FastAPI - Uvicorn | Jenkins (tools for software configuration management) | PHP | CSS | Web Development | Javascript (React.js) | IBM Data stage | PostgreSQL | JavaScript (jQuery, Node.js, React Native) | React.js, Node.js, Bootstrap, HTML, CSS | SAP Migration | Cloud platform experience (MS Azure, Google Cloud, AWS) | LLM | RAG | Vector Databases, AI Agents, Agentic AI, LLM Evaluation, AI Orchestration